

Automobile Engineering • Semester II

# Basic Electrical and Electronics Engineering

Course Code: 2051

Credits: 4

Revision: 2026

Theory

Fundamental principles of DC/AC circuits, electrical machines, automotive power sources, and electronic automation.

[Download Study Notes \(PDF\)](#)

## Official Source Declaration

This lesson is strictly based on the **SITTTR Kerala Revision 2026 Syllabus** for Course Code **2051** (Basic Electrical and Electronics Engineering). All modules, learning outcomes, instructional hours, and assessment patterns are extracted directly from the official PDF.

- **Program:** Automobile Engineering
- **Course Type:** Theory (L:4, T:0, P:0, S:0)
- **Total Hours:** 60 Instructional + 90 Self-learning

## Course Dashboard

### Teaching Scheme

| Component          | Hours/Week |
|--------------------|------------|
| Lecture (L)        | 4          |
| Self-Learning (SS) | 6          |

|                      |                    |
|----------------------|--------------------|
| Total Semester Hours | 60 (Instructional) |
|----------------------|--------------------|

## Prerequisites

- Basic Science - Physics (2002A)
- Basic Mathematics (1002)

## How to Study this Lesson

This course is designed for Automobile Engineering students. Focus on how electrical principles apply to vehicle systems like charging, ignition, and ECUs.

**Study Tip:** Use the interactive Ohm's Law calculator in Module 1 and the Transformer EMF calculator in Module 2 to practice numerical problems.

## Course Outcomes (CO)

| CO No. | Outcome Description                                                                                                       | Bloom's Level |
|--------|---------------------------------------------------------------------------------------------------------------------------|---------------|
| CO1    | Explain fundamental principles of DC/AC circuits, electromagnetic induction, and 3-phase systems to solve basic problems. | Understand    |
| CO2    | Describe construction, working principles, and applications of DC motors, AC induction motors, and transformers.          | Understand    |
| CO3    | Discuss working of automotive power sources (batteries, generators, alternators) and industrial heating.                  | Understand    |
| CO4    | Explain electronic switching devices (MOSFET, SCR, Transistors), logic gates, and automotive automation architecture.     | Understand    |

## Syllabus Coverage

| Module | Title               | Instructional Hours | Weightage |
|--------|---------------------|---------------------|-----------|
| I      | Electrical Circuits | 15                  | 25%       |

| Module | Title                                      | Instructional Hours | Weightage |
|--------|--------------------------------------------|---------------------|-----------|
| II     | Electric Motors and Transformers           | 15                  | 25%       |
| III    | Electrical Sources and Power Utilisation   | 15                  | 25%       |
| IV     | Fundamentals of Electronics and Automation | 15                  | 25%       |

## Learning Roadmap

### Step 1: Fundamentals (Module I)

Master Voltage, Current, and Ohm's Law. Understand how AC and 3-phase power works.

### Step 2: Conversion (Module II)

Learn how electrical energy is converted to mechanical energy (Motors) and how voltage levels are changed (Transformers).

### Step 3: Generation & Storage (Module III)

Study how cars generate power (Alternators) and store it (Batteries).

### Step 4: Control (Module IV)

Understand the electronic "brain" of the vehicle using transistors, logic gates, and sensors.

Module I: Electrical Circuits

15 Hours



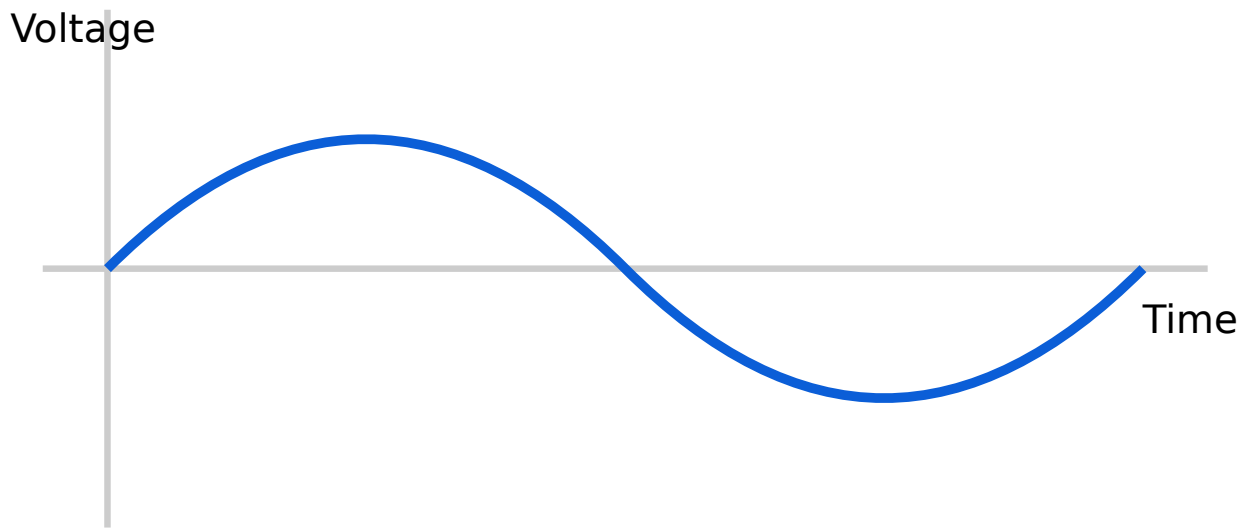


Fig 1.1: Sinusoidal AC Waveform

### Key AC Terms:

- **RMS Value:** The effective value of AC.  $V_{rms} = V_{max} / \sqrt{2}$ .
- **Frequency (f):** Number of cycles per second. Unit: Hertz (Hz).
- **Power Factor:** Ratio of real power to apparent power.

## Module II: Electric Motors and Transformers

15  
Hours

### 2.1 DC Motors

A DC motor converts DC electrical energy into mechanical energy. It works on the principle that a current-carrying conductor placed in a magnetic field experiences a force.

Types of DC Motors in Automobiles

| Type         | Application                          |
|--------------|--------------------------------------|
| Series Motor | Starter Motor (High starting torque) |
| Shunt Motor  | Wiper motors, cooling fans           |

## 2.2 Transformers

A static device that transfers electrical energy between circuits through electromagnetic induction, usually changing voltage levels.

### EMF Equation:

$$E = 4.44 \times f \times \Phi_m \times N$$

$f$  = frequency,  $\Phi_m$  = max flux,  $N$  = number of turns

**Turns Ratio (K):**  $V_2/V_1 = N_2/N_1 = I_1/I_2$ .

### Malayalam support:

ഏകദേശം ഏകദേശം ഏകദേശം (Step-up) ഏകദേശം ഏകദേശം (Step-down) ഏകദേശം ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം (Ignition Coil) ഏകദേശം ഏകദേശം-ഏകദേശം ഏകദേശം ഏകദേശം.

## Module III: Electrical Sources and Power Utilisation

15  
Hours

### 3.1 DC Generator & Alternator

In modern cars, the **Alternator** is used to charge the battery and power electrical systems while the engine is running. It produces AC, which is then rectified to DC.

### 3.2 Batteries

Automobiles primarily use **Lead-Acid Batteries**. They consist of lead plates in an electrolyte of sulfuric acid.

- **Lead-Acid:** Reliable, high cold-cranking amps.
- **Alkaline (Nickel-Iron):** Robust, long life, but lower efficiency.
- **VRLA:** Valve Regulated Lead Acid (Maintenance-free).

### 3.3 Electric Heating

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**Joule's Law:**  $H = I^2 R t$ . Heat produced is proportional to the square of current, resistance, and time.

- **Induction Heating:** Used for melting metals (Eddy currents).
- **Dielectric Heating:** Used for non-conductors (plastics, textiles).

## Module IV: Fundamentals of Electronics and Automation

15  
Hours

### 4.1 Diodes and Rectifiers

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A **PN Junction Diode** allows current in one direction only (Forward Bias).

**Rectifiers** convert AC to DC.

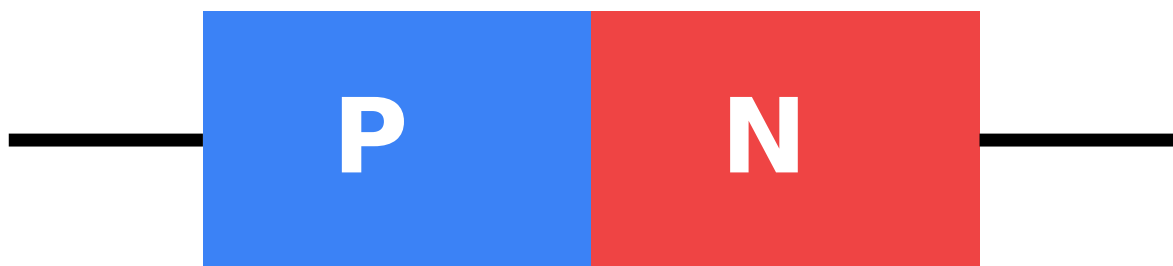


Fig 4.1: PN Junction Diode Structure

### 4.2 Transistors and MOSFETs

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Transistors act as **switches** or **amplifiers**. In automotive ECUs, Power MOSFETs are used for high-speed switching of fuel injectors and ignition coils.

### 4.3 Logic Gates & Automation

Logic gates (AND, OR, NOT, NAND, NOR) form the basis of digital control. **Universal Gates:** NAND and NOR.

**Automotive Control System:** Sensors → ECU → Actuators.

Malayalam support:

സെൻസറുകൾ (Sensors) വാഹനത്തിൽ വിവിധ സ്ഥാനങ്ങളിൽ, ECU ന്നോട് ബന്ധിതമായി സ്ഥാപിക്കുന്നു, പ്രവർത്തനങ്ങൾ (Actuators) നിയന്ത്രിക്കുന്നു. ഉദാഹരണത്തിന്, താപനില സെൻസർ Thermistor ഉപയോഗിക്കുന്നു.

## Suggested Student Activities

| No | Activity Description                                                         |
|----|------------------------------------------------------------------------------|
| 1  | Identify battery, fuse box, and wiring harness in a vehicle.                 |
| 4  | Calculate total resistance and power of headlamp circuits (Series/Parallel). |
| 9  | Disassemble a used lead-acid battery cell safely and sketch components.      |
| 13 | Decode resistor color bands and verify with a multimeter.                    |

## Formula Bank

Ohm's Law

$$V = I \times R$$

Electrical Power

$$P = V \times I$$



Transformer Ratio

$$N1/N2 = V1/V2$$

Joule's Heating

$$H = I^2 R t$$

AC RMS Voltage

$$V_{rms} = V_{max} / 1.414$$

Induction Motor Slip

$$s = (N_s - N_r) / N_s$$

## Visual Library

### Circuit Symbols

Standard symbols for Resistors, Capacitors, Diodes, and Transistors used in automotive schematics.

### Motor Construction

Internal views of DC Shunt Motors and Squirrel Cage Induction Motors.

## Assessment Methodology

| Assessment     | Marks | Details                                            |
|----------------|-------|----------------------------------------------------|
| CIE (Internal) | 40    | Activities (15), Series Exams (20), Attendance (5) |

|                |    |                       |
|----------------|----|-----------------------|
| ESE (Semester) | 60 | 2.5 Hour Written Exam |
|----------------|----|-----------------------|

## ESE Question Paper Pattern

- **Part A:** 8 questions (2 per module) × 1 mark = 8 Marks
- **Part B:** 8 questions (2 out of 3 per module) × 3 marks = 24 Marks
- **Part C:** 4 questions (1 set per module with choice) × 7 marks = 28 Marks

## Module-wise Questions

### Module I

#### State Ohm's Law and its limitations.

Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided temperature and other physical conditions remain constant ( $V=IR$ ). Limitations: It does not apply to non-ohmic conductors (like semiconductors) or when temperature varies.

### Module II

#### Why is a DC series motor used as a starter motor in cars?

A DC series motor provides very high starting torque, which is necessary to overcome the inertia and compression of the internal combustion engine during startup.

### Module III

#### Explain the purpose of an alternator in a vehicle.

The alternator converts mechanical energy from the engine into AC electrical energy, which is rectified to DC to charge the battery and power the vehicle's electrical components (lights, ECU, etc.) while the engine is running.

### Module IV

## What are universal gates? Why are they called so?

NAND and NOR gates are called universal gates because any other logic gate (AND, OR, NOT) can be constructed using only these gates.

## Practice Model Paper

**Note:** This is a practice paper based on the official pattern, not an official examination paper.

### Part A (1 Mark each)

1. Define RMS value of AC.
2. What is the unit of electrical energy?
3. Define slip in an induction motor.
4. Mention one application of a transformer.

### Part B (3 Marks each)

1. Compare single-phase and three-phase systems.
2. Explain the working principle of a DC motor.
3. Differentiate between lead-acid and alkaline batteries.
4. Draw the V-I characteristics of a PN junction diode.

## Quick Revision

### Key Points

- $V=IR$ ,  $P=VI$ ,  $E=Pt$ .
- Transformers change voltage, not frequency.
- Alternators charge the battery in cars.
- Diodes = Rectifiers; Transistors = Switches.

### Exam Checklist

- [ ] Definitions of V, I, R, P, E.
- [ ] Faraday's Laws of Induction.

- [ ] DC Motor types and applications.
- [ ] Transformer EMF equation.
- [ ] Logic gate truth tables.

## Glossary

### **ECU**

Electronic Control Unit - The "computer" in a vehicle.

### **Rectifier**

A circuit that converts AC to DC.

### **Stator**

The stationary part of an electric motor/generator.

### **Rotor**

The rotating part of an electric motor/generator.

### **Semiconductor**

Material with conductivity between a conductor and insulator (e.g., Silicon).

## References

- Fundamentals of Electrical Engineering and Electronics - B.L. Theraja (S. Chand)
- Electronic Devices and Circuits - G.K. Mittal (Khanna Publishers)
- Automobile Electrical and Electronic Systems - Tom Denton (Routledge)
- [NPTEL: Basic Electrical Technology](#)